GlucoTwin Ward

A live glucose twin for every inpatient with diabetes: it warns about lows before they happen, checks insulin before it is given, targets monitoring and hands over new diabetes at discharge

[Team name] · Indian Institute of Management Lucknow · Digital Twin in Healthcare PoC Challenge 2026

Research prototype on synthetic patients · not a medical device · never recommends a dose

Diabetes on the ward goes wrong in predictable ways

~1 in 4

inpatients has diabetes or high glucose (27-hospital bedside audit)

1

Lows happen between checks

18% moderate and 2.8% severe lows in 54,502 admissions, linked to longer stays, readmission and 90-day mortality.

2

Insulin errors are routine

Up to 39% of insulin-treated inpatients had an insulin error; 29% of surgical patients had a peri-operative diabetes plan.

3

Steroid highs are missed by design

Morning steroids push glucose up in the afternoon and evening; wards check fasting glucose.

4

New diabetes leaves undiagnosed

About a quarter of hyperglycaemic admissions in a 2026 Delhi study had undiagnosed diabetes.

Sources: SHM Converge 2025 · Med J Aust 2021 (QuIDS) · Medicine Today 2014 · J Assoc Physicians India 2026

What research has not solved yet

Hospital risk models

Retrospective, single-centre, predict once at admission; not tested in real time (2025 studies).

Inpatient CGM

Cut serious lows in an RCT (Diabetes Care 2020), but not approved for hospital use in most places; needs staff and training.

Insulin dosing software

The one mature system is US-only. Indian wards run on paper sliding scales.

The gap

A live, patient-level picture of the next 12 hours, built from what the ward already records, that works on fingersticks today and uses a CGM when one is worn.

Four features, one screen each, at moments the ward already has

1 Ward risk board

Shift start

Every patient sorted by chance of a low or severe high in the next 12 h, with reasons from their own record. Red only for lows.

2 Before you give insulin

Before each dose

Bedside checks plus the chance of a low if given, reduced or held. Doctors can try a dose. Never proposes one.

3 Smart monitoring

Overnight

An extra fingerstick at the riskiest hour for high-risk patients; a CGM for the highest-risk ones.

4 Discharge handover

Going home

Glucose during the stay, lows, steroids, kidney changes, probable new diabetes, a follow-up to book.

Left out on purpose: chatbot · patient app for inpatients · automatic dosing

What makes it a twin

1. **Synchronise hourly**
2. **Use orders already written**
3. **Predict 12 h**
4. **Simulate a dose**
5. **Check before giving**
6. **Record decisions**

Known future

Tonight's basal, tomorrow's fasting order and the steroid course are written in advance. The twin uses them; published models cannot.

Honest uncertainty

Every forecast has an 80% range that contains the outcome 81-91% of the time on test patients.

Hybrid what-if

Insulin effects come from a pharmacology layer, validated against counterfactual simulations, because record-trained models cannot learn them.

Every decision recorded

Flags, what-ifs, given/held decisions, extra checks and bookings go to an audit log with the model version.

Synthetic ward, real hospital formats

800

admissions over 60 days: sepsis, cardiac surgery, COPD/asthma, heart failure, orthopaedics

73k

FHIR R4 resources (Bulk Data NDJSON): the only thing the twin reads

~50%

of admissions wear a CGM, which supplies outcome labels

5

process failures injected at stated rates (hidden ground truth)

Physiology

Meals and how much is eaten, rapid and basal insulin, sulfonylureas, steroids peaking in the afternoon, stress, kidney injury, counter-regulation.

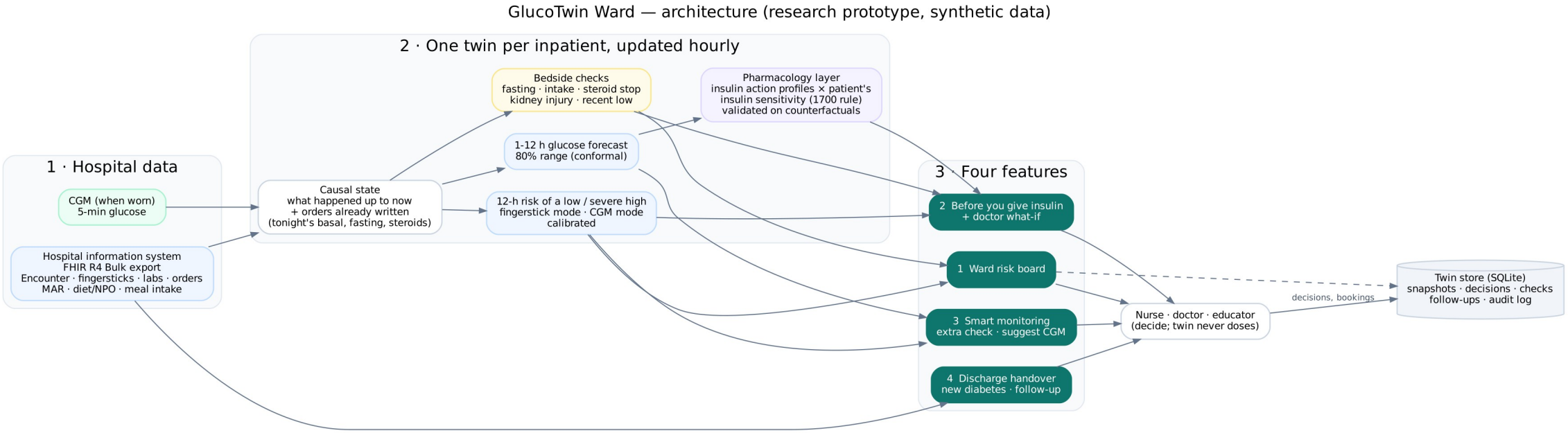
Ward process

Fingerstick protocol, sliding scale, 09:00 titration round, fasting orders for procedures, hypoglycaemia protocol with rechecks.

Counterfactual-ready

Each patient has independent random streams: change one dose and re-run, and everything before the dose is identical.

Architecture



Data in

FHIR R4 Bulk export from the hospital system + CGM. Swap synthetic for real without touching the models.

Models

LightGBM 12-h risk (fingerstick and CGM modes), 1-12 h forecasts with conformal ranges, pharmacology layer.

Out

Streamlit ward app and REST API; SQLite store with snapshots, decisions and audit.

The ward board at 21:00, before the bedtime insulin round



Ward board | Patient | Discharge | Evidence

Ward 7B · Medicine & Cardiac Step-down

Monday 24 August 2026 · 21:00 · 14 patients with diabetes or high glucose · risk over the next 12 hours

2 need attention | 5 watch | 0 extra check(s) suggested | 3 insulin dose(s) to review | 1 CGM suggested

	BED	PATIENT	LAST GLUCOSE	LOW RISK 12 H	SEVERE HIGH 12 H	WHY	NEXT CHECK	ACTION
red	7B-15	Rakesh Iyer · 70 Heart failure · day 3	117 1.5 h ago · fingerstick	>97%	2%	4 reading(s) under 100 in 24 h · Last fingerstick 117 mg/dL	22:00	insulin review; suggest CGM
red	7B-07	Tarun Pillai · 54 Heart failure · day 2	122 0 min ago · CGM	80%	<1%	3 reading(s) under 100 in 24 h · Went down to 68 at this time yesterday (CGM)	22:00	insulin review
amber	7B-22	Anil Patel · 61 Pneumonia / sepsis · day 8	308 1.5 h ago · fingerstick	<1%	>97%	Highest in 24 h: 308 mg/dL · Reached 328 yesterday afternoon	22:00	
amber	7B-02	Priya Joshi · 46 Cardiac surgery (CABG) · day 1	249 65 min ago · fingerstick	7%	56%	Highest in 24 h: 249 mg/dL	22:00	insulin review
amber	7B-21	Vikram Rao · 84 COPD / asthma exacerbation · day 6	208 1.5 h ago · fingerstick	2%	51%	Highest in 24 h: 238 mg/dL · Steroid 40 mg (prednisolone-equivalent) due: glucose usually peaks 2-8 pm	22:00	
amber	7B-06	Geeta Patel · 31 Cardiac surgery (CABG) · day 7	165 1.5 h ago · fingerstick	2%	51%	Highest in 24 h: 236 mg/dL · Reached 236 yesterday afternoon	22:00	
amber	7B-24	Ishita Verma · 65 Heart failure · day 6	184 1.5 h ago · fingerstick	9%	51%	Highest in 24 h: 220 mg/dL · Reached 208 yesterday afternoon	22:00	

1 Sorted by risk

Two red patients (risk of a low), five amber (likely severe high).

2 Reasons in ward language

From each patient's own record: readings under 100, steroid given this morning, fasting tomorrow.

3 Actions

Insulin doses to review; CGM suggested; extra check times.

One patient's twin: check, simulate, monitor



1

Before you give insulin

Basal 14 U due at 22:00: 98% chance of a low in 12 h if given as ordered.

2

Doctor what-if

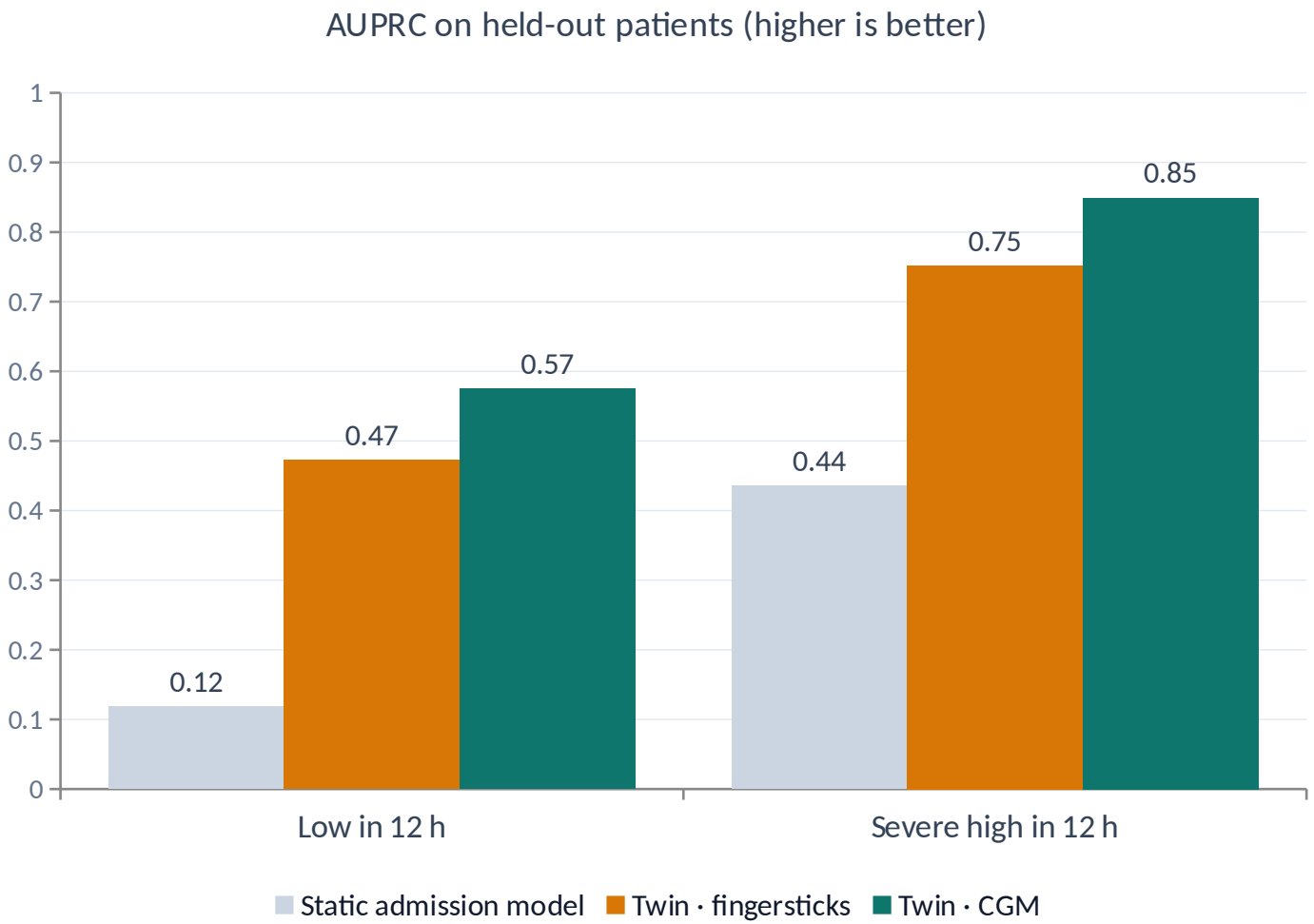
Hold the dose: forecast rises ~38 mg/dL (violet), chance of a low falls. The doctor decides; the decision is recorded.

3

Smart monitoring

Extra check suggested at 01:00 and a CGM for this patient.

The live twin beats an admission-time model by a wide margin



69%

of lows flagged red before they started (fingersticks), median 11 h ahead

0.64

red hours per patient-day: a light alert load

81-91%

of outcomes fall inside the twin's 80% forecast range

The insulin check catches unsafe doses before they are given

Injected process failure	Flagged before the dose
Mealtime insulin while nil by mouth	**24 / 24**
Basal not reduced the night before a fast	**38 / 38**
Insulin not reduced after the steroid course ended	**26 / 31**
Insulin not reduced after kidney injury	**81 / 91**
Sulfonylurea given while fasting	**7 / 7**

92%

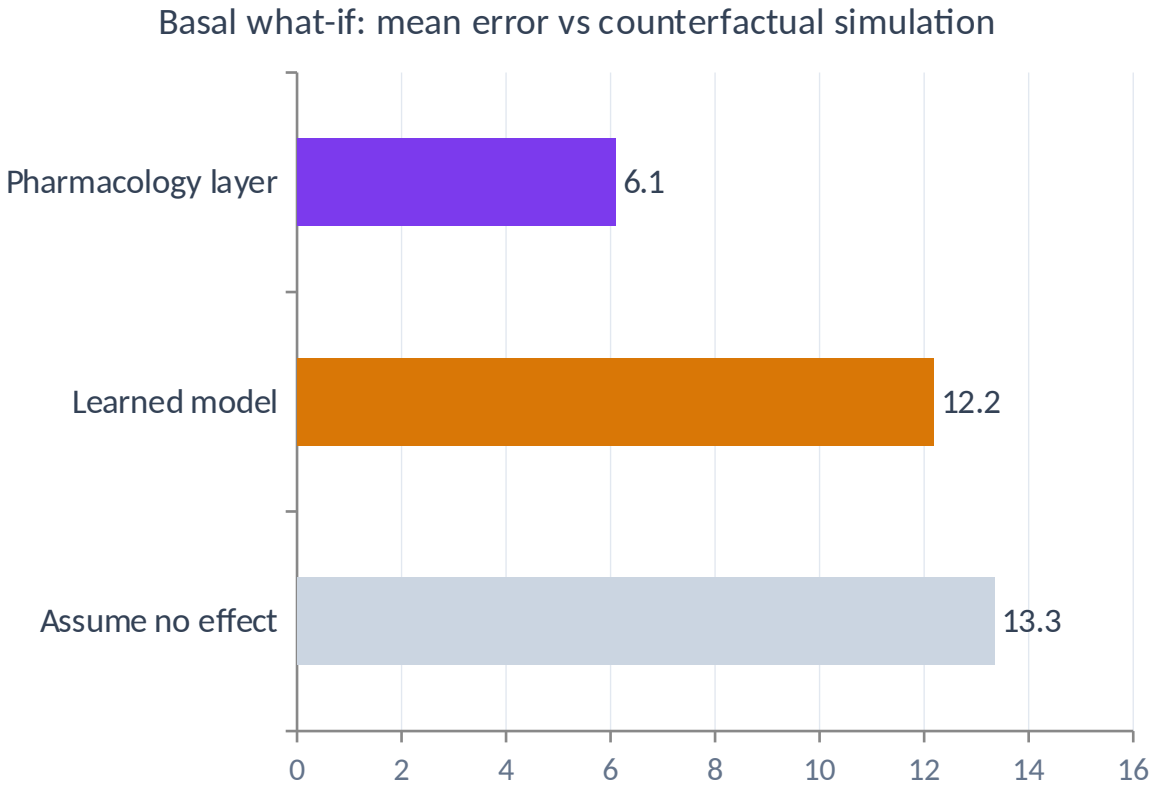
of all injected unsafe situations flagged (bedside rules, all 800 admissions)

4.7%

of routine doses get a high-level flag: low noise for nurses

Rules a nurse can verify at the bedside: fasting, poor intake, steroid stopped, kidney injury, low recent glucose. The twin never proposes a dose.

Dose what-ifs: pharmacology beats learning from records



Mealtime insulin: error 5.4 vs 10.7 mg/dL · 60 test dose events

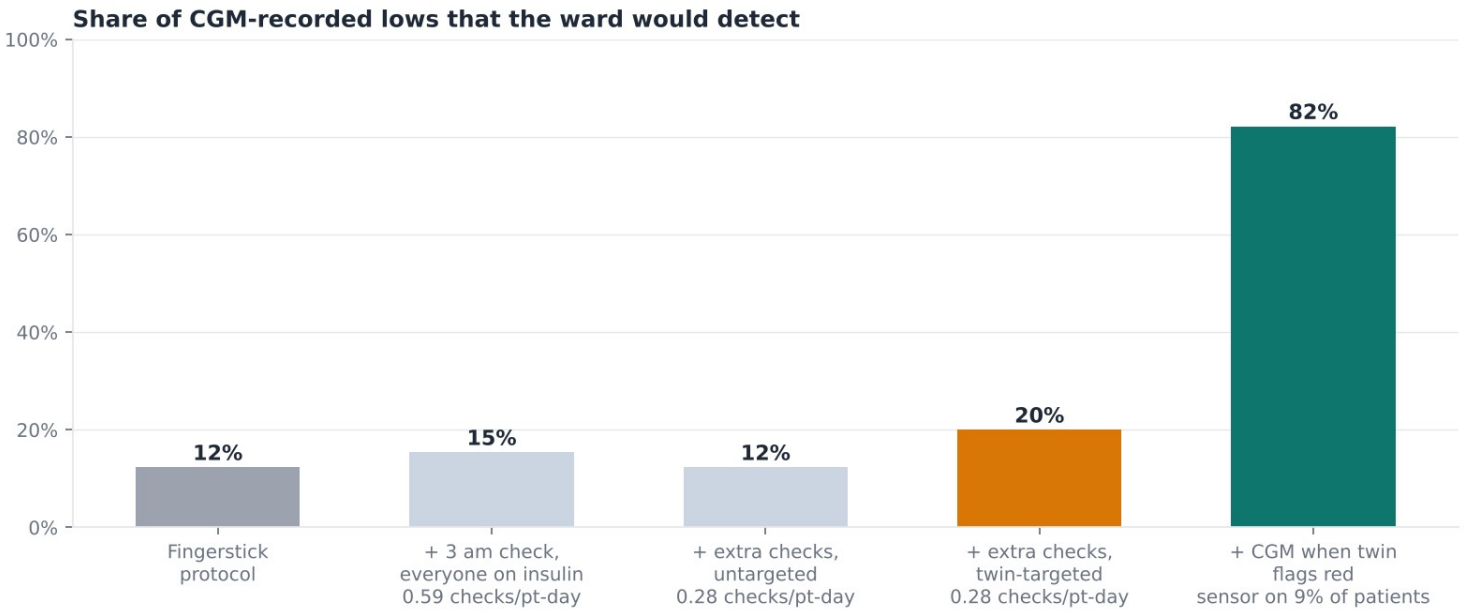
Holding tonight's basal: glucose at 6 h

True +25 · twin +23 · learned model +2 mg/dL

Why

Doctors give more insulin to patients who run high, so a model trained on records cannot separate a dose's effect from its reason (confounding by indication). The twin uses textbook insulin action profiles × the patient's insulin sensitivity, scaled on training patients and tested on counterfactual re-runs of test patients.

Fingersticks miss most lows: target the patients who need more



12%

of CGM-recorded lows are caught by routine fingersticks

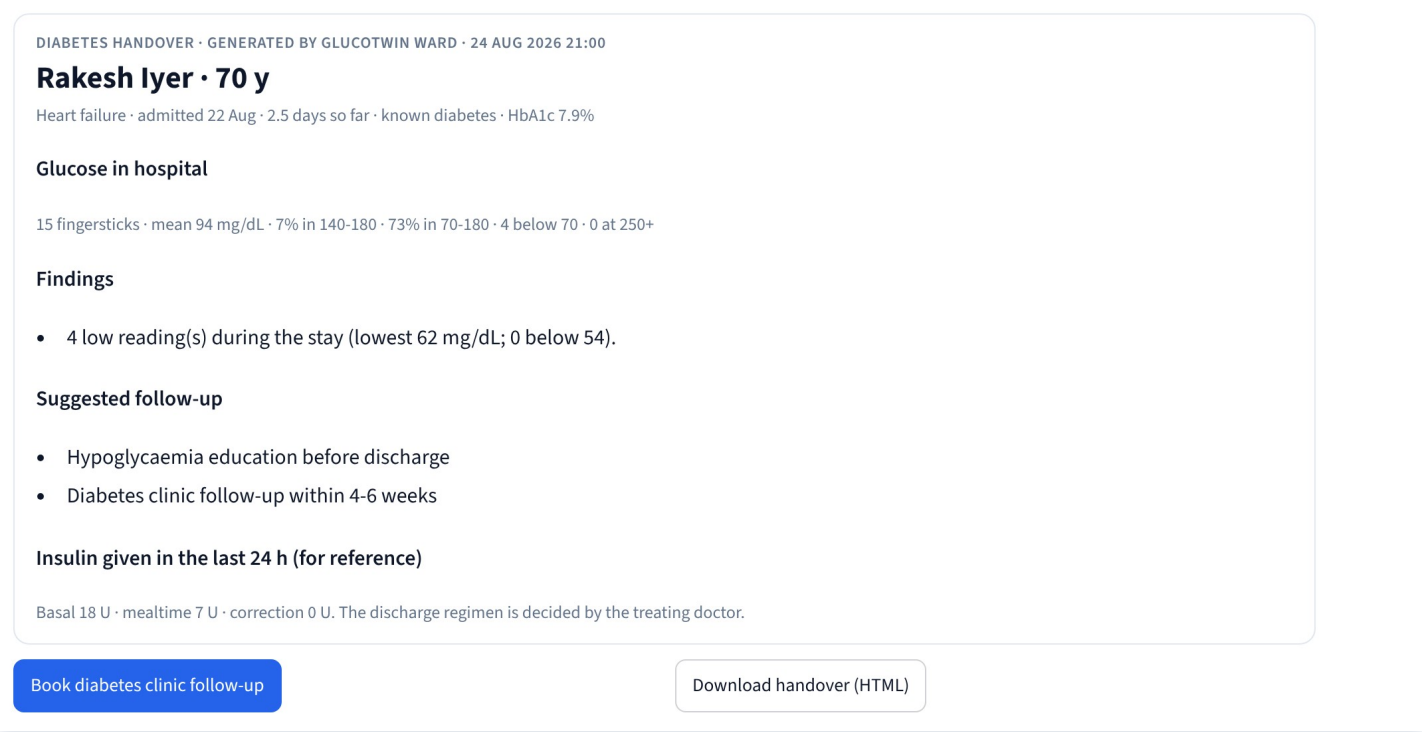
4.7 vs 0.9

extra lows caught per 100 extra checks: twin-targeted vs a 3 am check for everyone

80%

of lows happen after the twin's first red flag: a CGM then needs a sensor on only 9% of patients

Discharge handover closes the loop the ward drops



91%

of 127 patients with undiagnosed diabetes identified (64%) or prompted for HbA1c (28%)

0

false new diagnoses among patients with stress hyperglycaemia only

Also on the page: lows during the stay, steroid course, kidney changes, insulin started in hospital, a follow-up to book. It brings follow-up visits back to the hospital's own diabetes clinic.

From prototype to one real ward

1 Integrate

4-6 weeks

FHIR export from the hospital system: encounters, fingersticks, MAR, diet/NPO orders, labs. Connected glucometers if available.

2 Shadow mode

3 months

One ward. Predictions recorded, not shown. Measure real accuracy; recalibrate thresholds; test the what-if on real dose changes.

3 Live

3 months

Board and checks shown to staff; compare with shadow period and a matched ward.

4 Decide

Gate

Expand, change or stop on lows per 100 patient-days, good diabetes days, flags overridden, nurse time, new diabetes captured, follow-ups.

Regulation and privacy

CDSCO medical device software guidance (July 2026): prediction/monitoring software is medical device software; AI must disclose training data (real or synthetic), Indian sub-population robustness, change control and drift monitoring. Positioned as decision support with human review and no dose recommendation. DPDP by design.

Limitations, stated plainly

1. All evidence is synthetic: the next step is shadow mode on real ward data.
2. The simulator sets physiology, ward routines and failure rates; real wards will differ.
3. Dose what-ifs are validated on counterfactuals that share the simulator's assumptions.
4. Fingersticks have a low ceiling for catching lows; that is why the CGM suggestion matters.
5. Not covered: ICU, insulin infusions, tube feeds, type 1 diabetes, DKA, children, pregnancy.

Every inpatient with diabetes gets a twin

that warns, checks, targets and hands over.

0.47 vs 0.12
AUPRC for lows: twin vs admission model

92%
unsafe doses flagged before giving

80%
of lows covered by a CGM on 9% of patients

91%
undiagnosed diabetes caught at discharge

`github.com/ritwiksarkar11/glucotwin` · `make app`

Research prototype on synthetic patients · Apache-2.0